
EEEM071 Advanced Topics in Computer Vision and Deep Learning

Lab Weeks 5-10

Image Classification on FashionMNIST

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1 Introduction

Python is a widely-used, high-level, interpreted programming language known for its simplicity and versatility. It supports multiple programming paradigms and has a vast standard library.

Jupyter Notebook is an interactive programming environment that supports multiple programming languages and consists of cells that can contain code, text, or equations. It is widely used for data exploration and experimentation.

PyTorch is an open-source machine learning library for Python, primarily used for developing deep learning models. It provides a high-level interface and supports dynamic computational graphs for flexible and efficient model training.

Google Colab is a cloud-based service by Google that allows users to write and run Python code in a browser-based notebook environment. It provides a free computing environment with pre-installed popular libraries and supports collaborative work on Jupyter notebooks.

In this lab (from Week 5 to Week 10), you will learn how to complete a Deep Learning project with image classification on FashionMNIST as a starting point. We provide an off-the-shelf **Colab Notebook** for your reference. In the following weeks, we will explain every part of this implementation and show how to use Google Colab, Python, and PyTorch step by step.

It is **strongly encouraged** to get your hands dirty and pay more effort into this lab. This will help you a lot to finish the final coursework and MSc project.

2 Outline for Week 5 to Week 10

1. Week 5: Basic Python
2. Week 6: Numpy & PyTorch Tensor Operation
3. Week 7: PyTorch Dataset
4. Week 8: PyTorch Model Architecture
5. Week 9: PyTorch Optimization
6. Week 10: PyTorch Evaluation Metrics

For each week, we will release the lab sheet so that you can learn how to modify the module in the Colab Notebook and complete the coursework.

3 Image Classification with FashionMNIST

To begin working on a computer vision problem – image classification, let's get a computer vision dataset. We're going to start with FashionMNIST. MNIST stands for Modified National Institute of Standards and Technology. The **original MNIST dataset** contains thousands of examples of handwritten digits (from 0 to 9) and was used to build computer vision models to identify numbers for postal services. Amongst Deep Learning enthusiasts, solving MNIST is largely considered the equivalent to printing your first 'Hello World!' in a new programming language.

FashionMNIST, made by Zalando Research, is similar to MNIST but with data that is slightly harder to classify. It contains grayscale images of 10 different kinds of clothing items, as opposed to handwritten digits in MNIST.

You will be introduced to a variety of techniques in this notebook. First, you will learn how to download the data, prepare it, and conduct basic exploratory data analysis (EDA). Following that, there will be three neural networks to attempt to solve the problem of classifying the images in the dataset. With each network, a degree of complexity will be added. You will then conduct a basic evaluation and comparison of all the models you created. Finally, you will learn how to save your models and re-load them for future use.

4 Getting Started

Open this [link](#) and follow the steps in Figure 1 to 7 to copy the provided Colab file into your own Google drive and setup the environment.

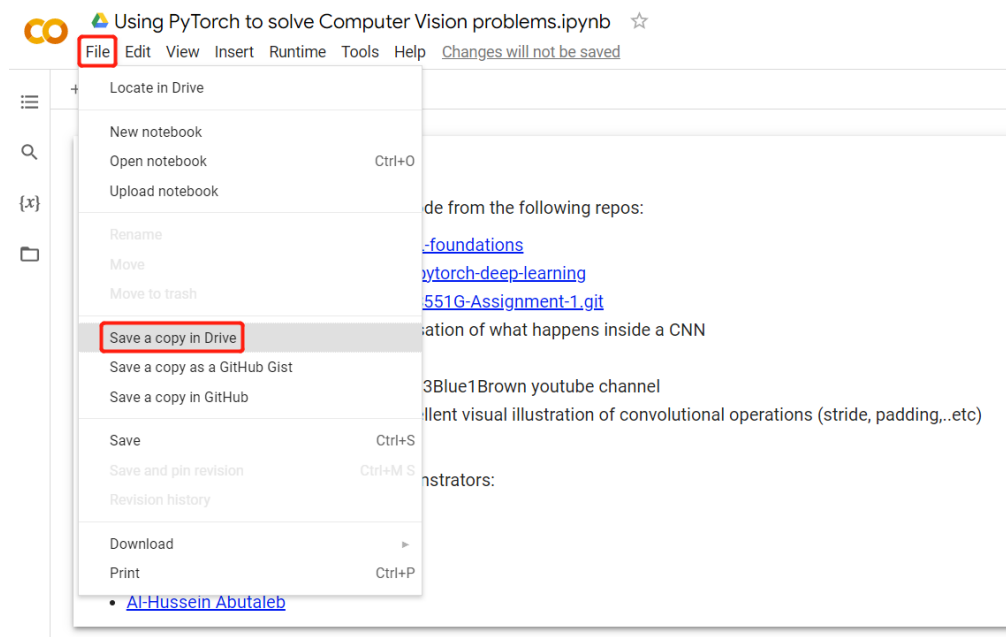


Figure 1: Copy to your Google drive

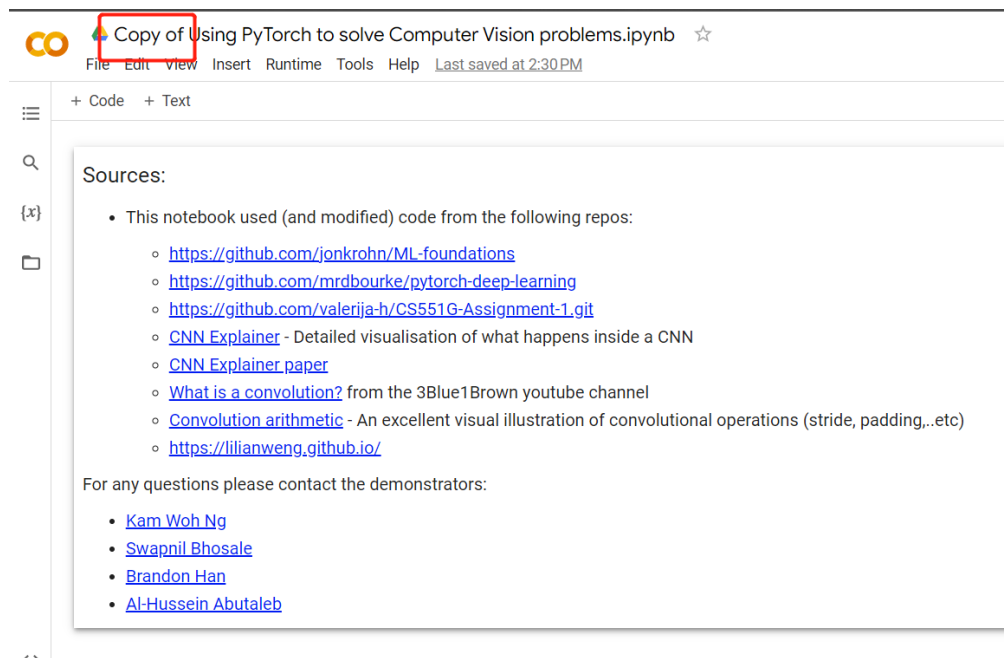


Figure 2: It will open a new link and the title should include the word “Copy of”. You can also rename your filename if you want to.

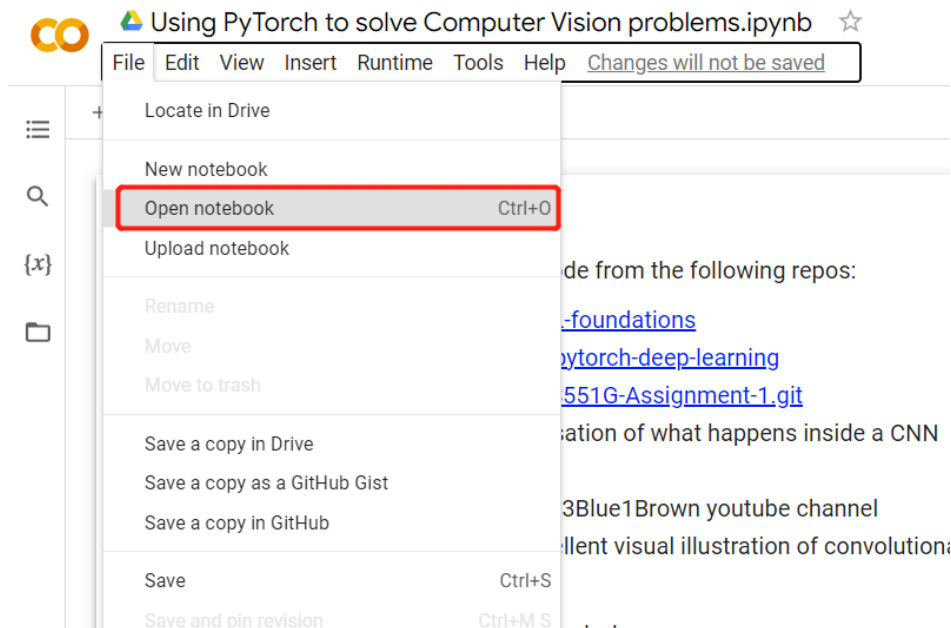


Figure 3: If you forgot the link to your own Colab file, you can click “Open notebook” in the original link.

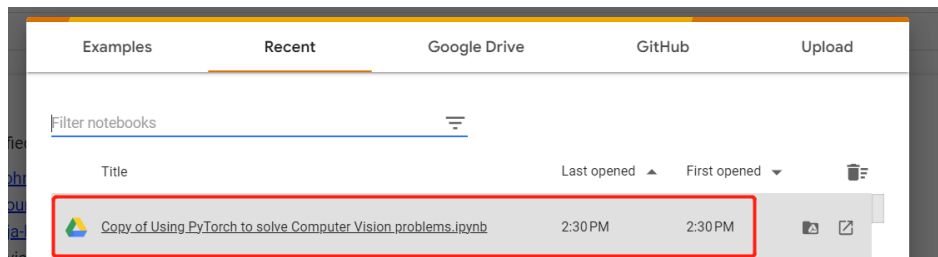


Figure 4: Choose the Colab file from the list.

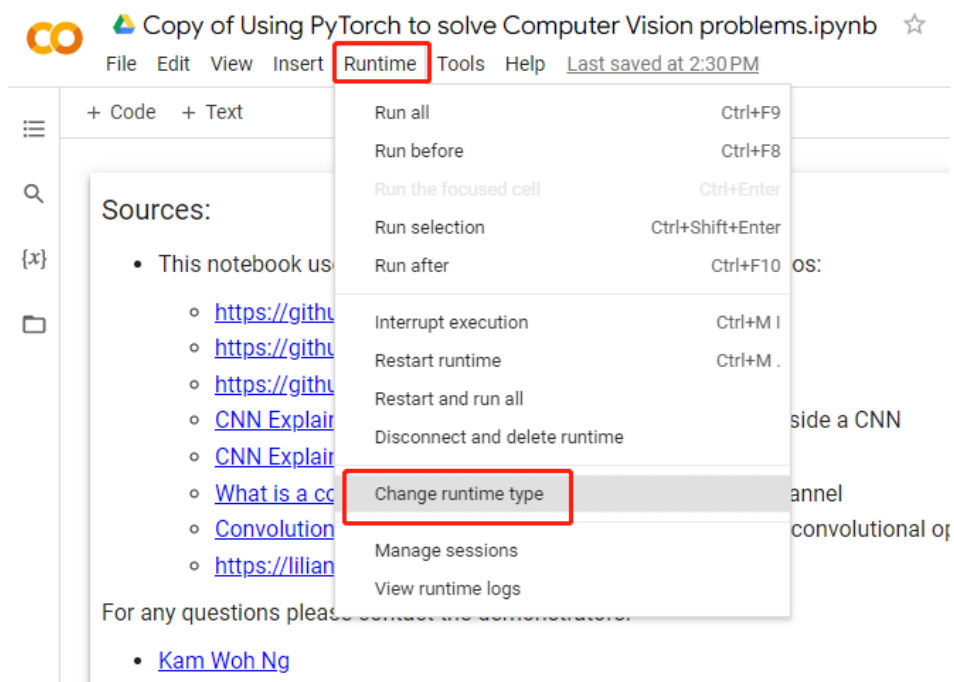


Figure 5: Choose runtime type.

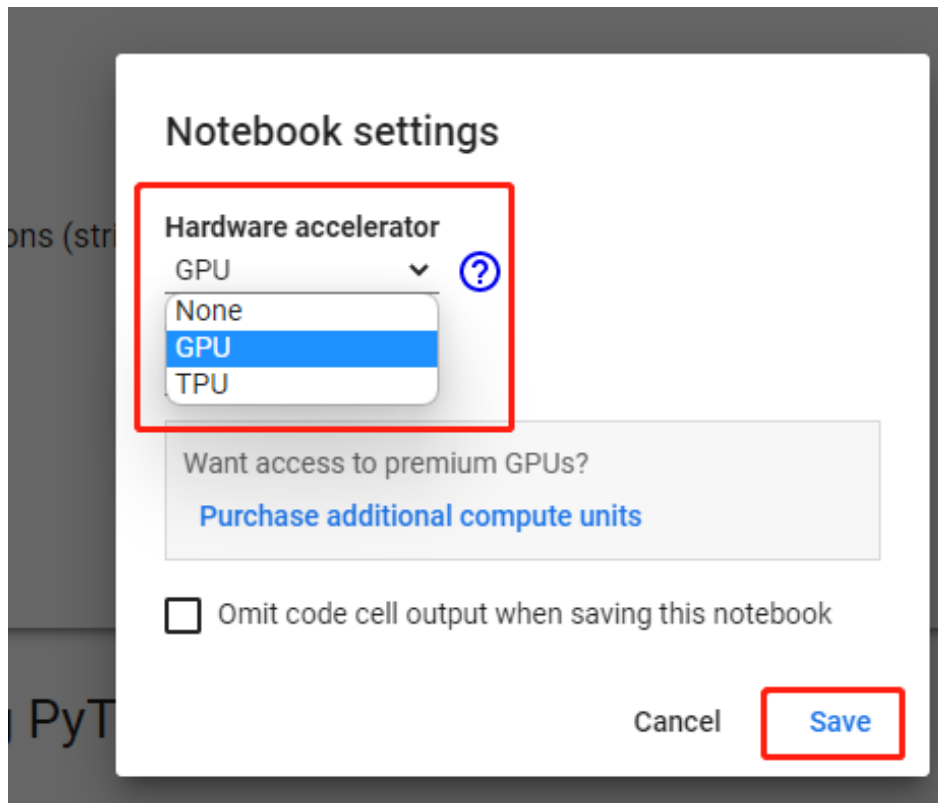


Figure 6: Ensure that the Colab environment is using GPU.



Figure 7: Familiarize yourself by clicking different buttons and seeing the output. To run a section of the code, you can click the “run” button at the left top corner of the code section. You can run each section and see the output.